

Implementation of DeFT: A Deadlock-Free and Fault-Tolerant Routing Algorithm for 2.5D Chiplet Networks

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Abstract—This project implements and evaluates a Noxim-based `DEFT_2_5D` simulator extension for studying DeFT-style routing in a 2.5D chiplet network. The implementation models four chiplets on an active interposer, a deterministic physical Vertical Link (VL) inventory, startup permanent VL faults, DeFT virtual-network assignment rules, movement-transition filtering, schema-v1 offline VL lookup tables, explicit XY baseline modes, synthetic traffic profiles, machine-readable metrics export, and traceable experiment and analysis helpers. The final fixed-window sweep completed 150 simulator runs across routing modes, traffic profiles, fault masks, and seeds. The generated fixed-window results are valid for descriptive report support only. The measured data set contains empty and partial injection cells, including 54 completed runs with zero packets injected after the warm-up boundary. Therefore, this report presents finite-window measurements with coverage counts and limitations, while using the T0033 diagnosis only to explain the XY blank-cell provenance and avoiding unsupported performance conclusions. A later opt-in drain-mode validation set completed all 95 documented DEFT cases, and a separate `INTERPOSER_AWARE_XY` versus DEFT drain comparison completed 68 matched rows in both modes while leaving 27 IA-XY timeout or non-100% rows as descriptive-only evidence.

Index Terms—2.5D integration, chiplet networks, Network-on-Chip, deadlock-free routing, fault tolerance, Noxim.

I. INTRODUCTION

2.5D chiplet integration connects multiple smaller dies through an active interposer. This architecture can improve modularity and yield, but it also makes inter-chiplet routing sensitive to VL availability and deadlock-avoidance constraints. DeFT is a routing approach proposed for this setting; it combines virtual-network assignment rules with fault-aware VL selection so inter-chiplet packets can be routed through the interposer while respecting deadlock-avoidance constraints [1].

The goal of this project is to extend the cycle-accurate Noxim simulator [2] so DeFT-style behavior can be studied in a reproducible workflow. The implementation focuses on a four-chiplet system with 4×4 chiplet-local meshes, a modeled active interposer layer, four physical bidirectional VLs per chiplet, permanent startup VL faults, two virtual channels for VN.0 and VN.1, and descriptive final metrics for reachability, average latency, and network throughput.

Assumption: The current implementation models 16 physical bidirectional VLs and uses the final-sweep policy recorded in the project documentation for reporting fault rates. This

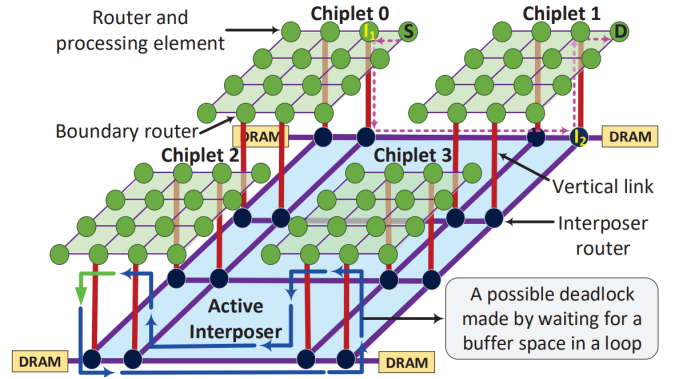


Fig. 1. Baseline 2.5D network architecture reused from the extended proposal source template.

differs from single-direction endpoint fault modeling and is preserved as a limitation.

II. RELATED WORK

Routing for chiplet-based and interposer-based systems has been studied through several deadlock-avoidance and reliability mechanisms. Modular turn restriction approaches such as MTR restrict dependency cycles in chiplet systems [3]. Remote-control and update-packet bypassing ideas explore alternate mechanisms for modular systems and 2.5D deadlock handling [4], [5]. ReD extends the virtual-network reliability direction for 2.5D interposer systems [6]. Fault-tolerant 3D NoC routing and configurable inter-die links provide related reliability context, but their assumptions differ from the up-down-up communication pattern of active-interposer 2.5D systems [7]–[9].

This project uses the original DeFT paper as the primary algorithmic reference for virtual-network assignment, movement restrictions, and design-time VL selection [1]. The implementation does not claim a new formal proof; it implements and validates the documented project surfaces inside Noxim.

III. PROBLEM DEFINITION

The project problem is to support deadlock-free and fault-tolerant inter-chiplet routing in a modeled 2.5D system while preserving traceable simulator behavior. The target topology

contains four CPU chiplets integrated on an active interposer. Each chiplet contains a 4×4 mesh, and each chiplet has four physical bidirectional VLs located at boundary routers.

The evaluation target from the project requirements is synthetic traffic under permanent VL fault scenarios, using reachability, latency, and throughput metrics. The implemented synthetic profiles are uniform, localized with 40% same-chiplet traffic probability, and hotspot traffic with three deterministic hotspot routers. Real-application PARSEC/GEM5 trace evaluation remains outside the current validated artifact set; PARSEC is a benchmark suite reference point, but no PARSEC/GEM5 trace workflow is validated in this final report [10].

Blocked: Real-application PARSEC/GEM5 trace evaluation would require a separate documented setup and validation task before inclusion.

IV. SYSTEM ARCHITECTURE AND DESIGN

A. Topology Model

The DEFT_2_5D topology uses a 2×2 chiplet grid. Chiplet router IDs are 0..63; interposer router IDs are 64..127. Packet sources and final destinations remain chiplet routers only. The interposer routers are internal transit routers.

Each chiplet owns four physical bidirectional VLs at deterministic boundary routers. The current model therefore has 16 physical bidirectional VLs. These physical links can also be described as 32 directional-equivalent channels for paper-aligned fault-rate reporting, but the simulator fault model disables one physical bidirectional VL as one object.

B. Virtual Networks

For DEFT_2_5D, VN.0 maps to VC 0 and VN.1 maps to VC 1. This avoids a parallel virtual-network field and keeps the routing state aligned with the existing Noxim virtual-channel flow-control path. Boundary-router reassignment is output-VC-aware so a head flit can reserve and forward into the selected VN safely.

The implemented movement filter enforces the current VN transition restrictions: VN.1 to VN.0 is rejected, Up-to-Horizontal movement in VN.0 is rejected, and Horizontal-to-Down movement in VN.1 is rejected. Because explicit Up and Down ports do not exist in the original Noxim direction set, the current implementation derives movement context from the layer, boundary-router state, input direction, and output direction.

C. VL Lookup Tables

The offline LUT generator emits a versioned `deft_vl_lut.v1` YAML format. Runtime DeFT routing loads the schema-v1 LUT, derives the active physical fault mask from current VL state, and selects source-side exit and destination-side entry VL records by lookup. Missing entries or nonfunctional selected VLs fail closed by returning no legal output direction.

V. IMPLEMENTATION

The project was implemented task by task and recorded in `docs/TASKS.md`, `docs/PROGRESS.md`, `docs/ARCHITECTURE.md`, and `docs/DECISIONS.md`. Main implemented surfaces are:

- `external/noxim/src/DeftTopology.*`: layer-aware router mapping, physical VL records, boundary-router inventory, and topology validation.
- `external/noxim/src/DeftFaultInjectionManager.*`: startup permanent physical VL fault configuration and validation.
- `external/noxim/src/DeftVirtualNetwork.*`: VN/VC mapping helpers and DeFT VN assignment support.
- `external/noxim/src/routingAlgorithms/Routing_DEFT.*`: runtime DeFT route selection through schema-v1 VL LUT entries.
- `external/noxim/src/DeftVerticalLinkLut.*`: schema-v1 VL LUT loader and validator.
- `external/noxim/config_examples/`: DEFT_2_5D topology, XY baselines, and synthetic traffic configurations.
- `external/noxim/other/deft_vl_lut_generator.py`: deterministic offline VL LUT generator.
- `external/noxim/other/deft_experiment_runner.py`: traceable experiment launcher.
- `external/noxim/other/deft_analysis_artifacts.py`: mechanical final-analysis artifact generator.
- `external/noxim/src/GlobalStats.*` and `external/noxim/src/ProcessingElement.*`: machine-readable metrics export and injected-packet counters.

Assumption: Generated artifacts under `external/noxim/other/generated/` are report-support outputs and remain traceable to their manifests.

VI. EXPERIMENTAL SETUP

The final sweep policy was defined before running the final matrix and was executed without changing simulator behavior during report drafting. Table I summarizes the matrix. The final sweep contains exactly 150 planned and completed simulator runs. No post-injection drain phase is included in the current fixed-window policy.

T0033 diagnosed the current standard XY implementation as a limited control baseline on DEFT_2_5D, not as an interposer-aware unrestricted inter-chiplet baseline. The existing XY route path selects only cardinal directions from the global footprint. It does not select VL, hub, interposer, or chiplet-layer phases. On DEFT_2_5D, unrestricted inter-chiplet routes require VL/hub/interposer traversal, so standard XY can select chiplet-layer movement that has no physical cross-chiplet cardinal link.

TABLE I
FINAL SWEEP MATRIX

Dimension	Values
Routing modes	XY, DEFT
Traffic profiles	uniform, localized_40, hotspot_3x10
Fault masks	0x0000, 0x0001, 0x0011, 0x0111, 0x1111
Physical fault rates	0%, 6.25%, 12.5%, 18.75%, 25%
Seeds	0, 1, 2, 3, 4
Simulation window	-sim 10000
Warm-up window	-warmup 1000
Stats format	JSON

The later drain-mode comparison therefore uses `INTERPOSER_AWARE_XY`, a separate baseline that can traverse the modeled interposer path. This baseline is not standard XY, and this report keeps those labels separate.

T0033 also showed that a post-injection drain/source-cutoff policy would be needed for eventual-delivery analysis. That policy would not by itself fix standard XY topology incompatibility on unrestricted inter-chiplet traffic, because standard XY still has no VL/hub/interposer route phase.

T0056 and T0053 later used the opt-in source-cutoff plus drain mode for deterministic fixture matrices. T0056 used DEFT only. T0053 used DEFT and `INTERPOSER_AWARE_XY` only, with the same 19 fixture definitions as T0056. These artifacts are not fixed-window final sweeps and are not substitutes for the T0026/T0027/T0028 tables.

Assumption: A zero-injection run is a completed simulator run with `total_injected_packets == 0` in the measured stats window, not a failed run.

Assumption: T0033 warm-up-0 diagnostics are traceability evidence that the traffic sources were not empty. They are not a replacement final performance data set.

Blocked: The historical T0026/T0027/T0028 fixed-window runner policy does not evaluate eventual delivery after a post-injection drain phase.

Blocked: The current physical bidirectional fault model does not represent the original paper’s single-direction 3.125% fault case.

VII. VALIDATION PROVENANCE

The final report-support results are traceable through the following generated artifacts:

- T0026 final sweep output: `external/noxim/other/generated/t0026_final_sweep_v1/`.
- T0026 final analysis output: `external/noxim/other/generated/t0026_final_analysis_v1/`.
- T0027 report-support output: `external/noxim/other/generated/t0027_report_support_v1/`.
- T0028 claim-safe results draft: `external/noxim/other/generated/t0028_final_report_results_v1/report_results_draft.md`.

T0026 validation established that the executed manifest had `mode: execute, run_count: 150, 150 completed runs, and 150 return code 0 runs`. All 150 JSON stats files existed and contained the T0020 metric fields. T0027 validation derived blank-aware condition and pair tables from T0026 raw artifacts and cross-checked them against the T0026 sweep summary, analysis run summary, and analysis comparison summary. T0028 converted the T0027 report-support tables into claim-safe prose and Markdown tables without rerunning simulations. The T0028 manifest keeps `claims_allowed: false`.

T0033 is additional diagnosis provenance only. It explains the source of the XY blank cells by inspecting the runner warm-up policy, injected-packet accounting, standard XY routing, and DEFT_2_5D topology wiring. It does not add new final performance rows, does not replace T0026/T0027/T0028, and does not change simulator behavior.

Additional post-fix drain-mode provenance is traceable through T0056 at `external/noxim/other/generated/t0056_deflt_post_fix_reachability_v1/` and T0053 at `external/noxim/other/generated/t0053_drain_iaxy_deflt_comparison_v1/`. T0056 validation established a 95-row DEFT drain-mode matrix over 19 deterministic fixtures and the accepted physical fault-mask ladder. All rows returned code 0, stopped with `drain_completed`, and passed with exact measured packet/flit delivery. T0053 validation established a 190-row drain-mode comparison matrix over DEFT and `INTERPOSER_AWARE_XY`. All DEFT rows passed. IA-XY passed 68 rows and timed out in 27 rows. The denominator-safe comparison table marks 68 matched rows as `complete_delivery_both_modes` and 27 rows as `descriptive_only_timeout_or_non100`.

VIII. RESULTS AND EVALUATION

The results in this section are assembled from the reviewed Markdown draft and the T0028/T0027/T0026 report-support artifacts. They are finite-window descriptive measurements. Blank cells are intentional and indicate that the supporting denominator is absent. The T0026/T0027/T0028 artifact chain remains the only final report-support data set used in the fixed-window tables below. T0056/T0053 drain-mode artifacts are summarized separately in Section VIII-F.

All 150 planned simulator runs completed with return code 0, and the generated T0027 report-support tables were cross-checked against the T0026 executed manifest, raw JSON stats, sweep summary, analysis run summary, and analysis comparison summary with zero mismatches. The upstream report-support manifest keeps `claims_allowed: false`.

The measured data set contains completed runs in which no packets were injected after the warm-up boundary. T0027 records 54 individual zero-injection runs. At condition level, 12 cells have packet injection in all five seeds, 13 cells have packet injection in only some seeds, and 5 cells have no packet injection in any seed. The 5 empty-injection cells are all XY with `hotspot_3x10` traffic.

TABLE II
ARTIFACT AND READINESS SUMMARY

Item	Value
Planned and completed runs	150
Raw stats rows	150
Condition cells	30
XY/DEFT pair rows	15
Individual zero-injection runs	54
Complete-injection condition cells	12
Partial-injection condition cells	13
Empty-injection condition cells	5
Cross-check mismatches	0
Claim state	claims_allowed: false

TABLE III
COVERAGE BY ROUTING AND TRAFFIC

Routing	Traffic	Runs	Nonempty	Zero	Injected	Received
DEFT	hotspot_3x10	25	22	3	336	134
DEFT	localized_40	25	25	0	1016	542
DEFT	uniform	25	24	1	436	238
XY	hotspot_3x10	25	0	25	0	0
XY	localized_40	25	20	5	35	0
XY	uniform	25	5	20	5	0

Under this fixed-window measurement policy, the XY hotspot cells cannot be compared with DEFT because the XY side injected zero packets in every hotspot fault-mask cell. T0033 diagnosed those XY|hotspot_3x10 zero-injection cells as measured-window artifacts: with -warmup 1000, early XY traffic had already injected before the measured window and then stalled on topology-incompatible paths. The warm-up-0 diagnostic values from T0033 are kept only as traceability evidence and are not reported as final performance results.

The XY uniform and localized cells injected packets in some seeds, but received zero packets in the measured window. T0033 diagnosed these XY|uniform and XY|localized_40 zero-received cells as standard XY topology incompatibility on DEFT_2_5D: cardinal-only XY cannot route unrestricted inter-chiplet traffic through the VL/hub/interposer path required by this topology. Therefore, side-by-side latency comparisons against DEFT are not supported.

A. Artifact Readiness

B. Coverage

C. Condition-Level Metrics

Reachability is $\text{total_received_packets} / \text{total_injected_packets}$ within the condition. Latency is received-packet-weighted across runs with received packets. Mean network throughput is the exported finite-window mean across all five runs. Status codes are: C = complete_injection_cell, P = partial_injection_cell, and E = empty_injection_cell. Blank reachability means no injected-packet denominator exists. Blank latency means no received-packet delay samples exist.

D. Pairwise Readiness

No delta columns are included. Table V describes whether values can be displayed side by side, not whether either routing mode should be preferred. The readiness code XY-empty means the XY side injected zero packets in the measured cell. The readiness code latency-blank means both modes have injection, but one mode has zero received packets, so latency comparison is blank.

E. Zero-Injection Runs

The full 54-run list is preserved in external/noxim/other/generated/t0027_report_support_v1/zero_injection_runs.csv. Table VI summarizes the zero-injection rows by routing and traffic profile.

F. Drain-Mode Post-Fix Validation and IA-XY Comparison

T0056 and T0053 use opt-in source-cutoff plus drain semantics rather than the T0026 fixed-window policy. These results are artifact-scoped and denominator-safe. They do not update the fixed-window condition metrics above and do not relabel IA-XY as standard XY.

The five T0056 all-pairs aggregate rows drained after 4088, 4858, 5617, 6239, and 5208 drain cycles for fault masks 0x0000, 0x0001, 0x0011, 0x0111, and 0x1111, respectively.

The denominator-safe T0053 comparison table classifies 68 matched fixture/fault rows as complete_delivery_both_modes and 27 rows as descriptive_only_timeout_or_non100. The latter rows are reported only as timeout/non-100% evidence; they are not used for ranking, improvement percentages, or statistical conclusions. For the IA-XY all-pairs aggregate rows, the five fault-mask cases stopped at drain_timeout after receiving 557, 1211, 888, 995, and 795 measured packets. These rows also retained source queues and in-network state at the timeout, so they remain descriptive-only.

IX. LIMITATIONS

Assumption: The final report text is descriptive support, not a new analysis layer.

Assumption: Aggregated reachability is blank when no packets were injected.

Assumption: Latency is blank when no packets were received.

Blocked: Empty or partial injection cells cannot support unqualified performance claims.

Blocked: XY|hotspot_3x10 cells injected zero packets for all seeds and fault masks in the measured window; T0033 diagnosed these cells as -warmup 1000 measured-window artifacts after early XY traffic injected and then stalled.

Blocked: The current data set does not support XY/DEFT latency comparisons because the XY side has zero received packets wherever it injected packets.

Blocked: Standard XY is cardinal-only on DEFT_2_5D and does not traverse VL/hub/interposer paths required for unrestricted inter-chiplet traffic.

TABLE IV
CONDITION-LEVEL DESCRIPTIVE METRICS

Routing	Traffic	Mask	Fault	Nonempty	Empty	Inj.	Recv.	Reach.	Latency	Thput.	Status
DEFT	hotspot_3x10	0x0000	0	5	0	69	19	0.27536232	12.73684211	0.0034222222	C
DEFT	hotspot_3x10	0x0001	6.25	4	1	19	4	0.21052632	16.75	0.0007111111	P
DEFT	hotspot_3x10	0x0011	12.5	3	2	114	72	0.63157895	40.36111111	0.0126888889	P
DEFT	hotspot_3x10	0x0111	18.75	5	0	66	18	0.27272727	7.61111111	0.0034666667	C
DEFT	hotspot_3x10	0x1111	25	5	0	68	21	0.30882353	9.47619048	0.0043333333	C
DEFT	localized_40	0x0000	0	5	0	289	166	0.57439446	23.37951807	0.0294888889	C
DEFT	localized_40	0x0001	6.25	5	0	144	72	0.5	16.83333333	0.0129777778	C
DEFT	localized_40	0x0011	12.5	5	0	114	60	0.52631579	16.08333333	0.0112	C
DEFT	localized_40	0x0111	18.75	5	0	303	155	0.51155116	18.50967742	0.0285111111	C
DEFT	localized_40	0x1111	25	5	0	166	89	0.53614458	18.43820225	0.0160888889	C
DEFT	uniform	0x0000	0	5	0	82	38	0.46341463	24.23684211	0.0071555556	C
DEFT	uniform	0x0001	6.25	5	0	137	84	0.61313869	23.96428571	0.0149777778	C
DEFT	uniform	0x0011	12.5	5	0	63	17	0.26984127	22.23529412	0.0032888889	C
DEFT	uniform	0x0111	18.75	5	0	81	49	0.60493827	20.08163265	0.0086	C
DEFT	uniform	0x1111	25	4	1	73	50	0.68493151	21.9	0.0087555556	P
XY	hotspot_3x10	0x0000	0	0	5	0	0				E
XY	hotspot_3x10	0x0001	6.25	0	5	0	0				E
XY	hotspot_3x10	0x0011	12.5	0	5	0	0				E
XY	hotspot_3x10	0x0111	18.75	0	5	0	0				E
XY	hotspot_3x10	0x1111	25	0	5	0	0				E
XY	localized_40	0x0000	0	4	1	7	0	0			P
XY	localized_40	0x0001	6.25	4	1	7	0	0			P
XY	localized_40	0x0011	12.5	4	1	7	0	0			P
XY	localized_40	0x0111	18.75	4	1	7	0	0			P
XY	localized_40	0x1111	25	4	1	7	0	0			P
XY	uniform	0x0000	0	1	4	1	0	0			P
XY	uniform	0x0001	6.25	1	4	1	0	0			P
XY	uniform	0x0011	12.5	1	4	1	0	0			P
XY	uniform	0x0111	18.75	1	4	1	0	0			P
XY	uniform	0x1111	25	1	4	1	0	0			P

TABLE V
XY/DEFT PAIRWISE READINESS

Traffic	Fault mask	XY status	DEFT status	Readiness
hotspot_3x10	0x0000	E	C	XY-empty
hotspot_3x10	0x0001	E	P	XY-empty
hotspot_3x10	0x0011	E	P	XY-empty
hotspot_3x10	0x0111	E	C	XY-empty
hotspot_3x10	0x1111	E	C	XY-empty
localized_40	0x0000	P	C	latency-blank
localized_40	0x0001	P	C	latency-blank
localized_40	0x0011	P	C	latency-blank
localized_40	0x0111	P	C	latency-blank
localized_40	0x1111	P	C	latency-blank
uniform	0x0000	P	C	latency-blank
uniform	0x0001	P	C	latency-blank
uniform	0x0011	P	C	latency-blank
uniform	0x0111	P	C	latency-blank
uniform	0x1111	P	P	latency-blank

TABLE VI
ZERO-INJECTION RUN SUMMARY

Routing	Traffic	Zero-injection runs
DEFT	hotspot_3x10	3
DEFT	uniform	1
XY	hotspot_3x10	25
XY	localized_40	5
XY	uniform	20

Blocked: The T0056/T0053 drain-mode artifacts are deterministic matrix evidence only; they do not prove universal reachability or universal IA-XY behavior.

Blocked: Standard XY remains cardinal-only. The drain-mode comparison uses INTERPOSER_AWARE_XY, which

TABLE VII
T0056 DEFT DRAIN-MODE SUMMARY

Item	Value
Fixtures	19
Fault masks	5
Simulator cases	95
DEFT drain-completed pass cases	95
DEFT timeout or non-100% cases	0
All-pairs aggregate rows	5
All-pairs packets per row	4032 / 4032
All-pairs flits per row	32256 / 32256

must not be relabeled as standard XY.

Blocked: Stronger result statements, non-empty XY hotspot measurements, latency comparisons, single-direction fault

TABLE VIII
T0053 IA-XY DRAIN-MODE OUTCOME SUMMARY

Fixture group	Rows	IA-XY pass	IA-XY timeout
route_family_pair	40	40	0
source_isolated	20	20	0
destination_stress	20	6	14
bounded_aggregate	5	1	4
bounded_low_load	5	1	4
all_pairs_aggregate	5	0	5
Total	95	68	27

cases, universal rankings, and improvement percentages require separate documented validation or rerun policy.

Blocked: PARSEC/GEM5 trace evaluation remains outside the current final-sweep artifact set.

X. REPRODUCIBILITY

The final LaTeX artifact is generated from docs/FINAL_REPORT_DRAFT.md, using Extended_Proposal.zip as the IEEE conference-style template reference. The final report source is intentionally separated from the extended proposal source so the original archive and proposal files remain unchanged.

The report-support data are traceable through task IDs T0025 through T0028. The T0033 diagnosis is provenance for interpreting blank XY cells, not a result source for the fixed-window tables. T0056 and T0053 are additional post-fix drain-mode artifact sets summarized separately. The simulator final sweep was not rerun during final-report conversion, T0035 report revision, or T0048 report integration. T0048 updates this LaTeX source only; PDF and package regeneration remain out of scope for that task.

XI. CONCLUSION AND FUTURE WORK

The project produced a traceable Noxim extension for DEFT_2_5D and a reproducible final-sweep artifact set. The final report-support data confirms that the planned 150-run matrix executed successfully and that generated summary tables cross-check against raw artifacts. The reportable fixed-window results should remain descriptive because the measured data contains blank and partial cells, and the known XY blank-cell behavior is now explained by the T0033 diagnosis.

The post-fix T0056/T0053 drain-mode artifacts add useful matrix-scoped validation evidence: DEFT completed the documented 95-case drain matrix, and IA-XY completed 68 of the 95 matched comparison rows while timing out in 27 rows. The safest conclusion is that the implementation and report-support workflow are in place, while stronger experimental claims require an explicitly scoped follow-up validation policy. Future work should address non-empty hotspot measurements for the standard XY control baseline, IA-XY timeout diagnosis, directional endpoint fault modeling, traffic-profile-specific LUT demand inputs, and validated PARSEC/GEM5 trace support before expanding the report claims.

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